

RESEARCH ARTICLE

Process Systems Engineering

AIChE
JOURNAL

A surrogate-based framework for feasibility-driven optimization of expensive simulations

Huayu Tian  | Marianthi G. Ierapetritou

Department of Chemical and Biomolecular Engineering, University of Delaware, Newark, Delaware, USA

Correspondence

Marianthi G. Ierapetritou, Department of Chemical and Biomolecular Engineering, University of Delaware, Newark, DE, USA.
Email: mgi@udel.edu

Funding information

US Food and Drug Administration (FDA), Grant/Award Number: FDABAA-20-00123

Abstract

Surrogate-based optimization approaches have been widely adopted in industrial problems due to their potential to reduce the number of simulation runs required in the optimization process. The surrogate-based optimization framework has been extended to feasibility analysis in pharmaceutical manufacturing to characterize the design space. Most surrogate-based approaches for feasibility analysis are limited to the construction of a regression model for the feasibility function. In this work, we developed a framework with the feasibility problem considered as a classification problem, and additional stages introduced to improve local exploitation and global exploration. We illustrate the efficiency of the proposed framework on three test problems and implement it in a realistic case study describing the production of solid-based drugs using wet granulation, aimed to reduce the operation cost, improve product quality, and increase process flexibility and robustness.

KEYWORDS

feasibility, Gaussian process, support vector machine, surrogate-based optimization

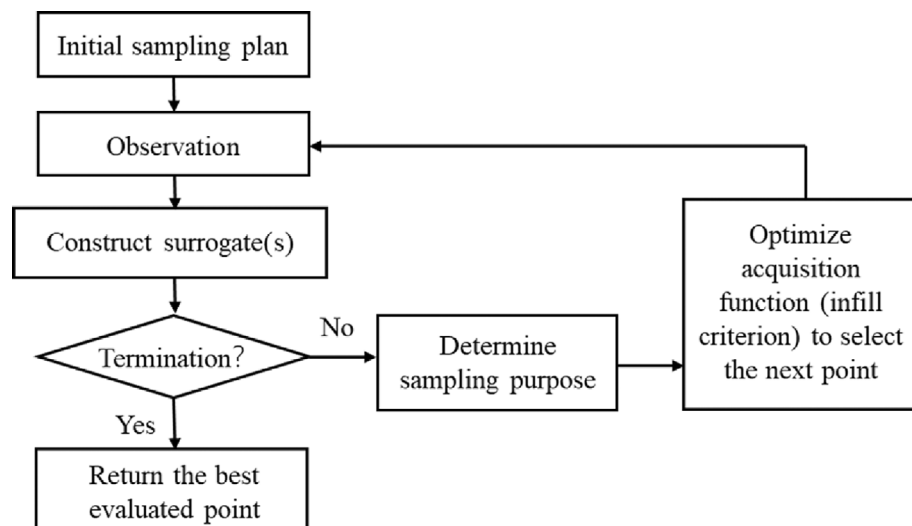
1 | INTRODUCTION

Simulation optimization (or simulation-based optimization) refers to the optimization of an objective function subject to constraints, both of which can be evaluated through a simulation.¹ The problem formulation, without loss of generality, can be represented as problem (Equation 1). The purpose is to find the optimal value of x , within the upper and lower bounds that can minimize the objective function $f(x)$, while satisfying a set of inequality constraints $g_i(x)$. Equality constraints can be transformed to two inequality constraints. Both $f(x)$ and $g_i(x)$ depend on the simulation model, which can make the functions expensive to calculate.

$$\begin{aligned} &\min f(x) \\ &\text{s.t. } g_i(x) \leq 0, i \in \{1, 2, \dots, I\} \\ &x_d^{lb} \leq x_d \leq x_d^{ub}, d \in \{1, 2, \dots, D\} \end{aligned} \quad (1)$$

Due to the fact that the process simulation is not explicitly available and expensive to evaluate, it is hard or computationally

expensive to use derivative-based methods to perform optimization. Therefore, derivative-free optimization, or optimization over black boxes techniques are often used for this type of problem. A review of algorithms and software implementations for derivative-free optimization is provided by Rios and Sahinidis.² Common methods include direct search methods,^{3–5} random search methods^{6,7} and model-based methods.^{8–10} Among the algorithms and methods, utilization of surrogate models (also known as metamodels) is a common way to extract the properties of the original function and guide the search. For example, derivative information and probability distribution function of the surrogate model prediction can be used to facilitate optimization. It also has the potential to reduce the number of simulations required in the optimization process. These benefits have attracted a lot of attention in the field of surrogate-based optimization (SBO) over the past decades. A basic framework for SBO is shown in Figure 1. After initial sampling, surrogate model(s) are built to approximate the input–output relationship. Then, an acquisition function (also known as infill criterion) is optimized to determine the next sample point. The simulation is run at the new added point and the surrogate model is

FIGURE 1 Basic framework of surrogate-based optimization.

updated based on the new input–output pair. This iterative procedure is called adaptive sampling, which plays a key role in SBO.

Among various SBO methods, the kriging-based optimization approach, also known as Bayesian optimization,¹¹ has gained an increasing popularity in different research communities. The kriging model has the ability to provide prediction of function values along with the prediction uncertainty by considering the prediction as a stochastic process. An expected improvement (EI) function has been proposed to utilize both information and serve as the infill criterion for iterative adaptive sampling in the efficient global optimization framework.¹⁰ This kriging-based optimization approach was first presented for unconstrained problems.¹⁰ For constrained problems, methods have been developed, which can be divided into two categories. The first category is to utilize an acquisition function that accounts for objective and constraints simultaneously. For example, Schonlau¹² suggested multiplying the value of the EI by the probability that the point is feasible. Following this idea, Gardner et al.¹³ rederived the acquisition function based on a proposed constrained improvement function as the improvement multiplied by feasibility indicator, and Gelbart et al.¹⁴ developed a more general framework that decoupled objective and constraint functions and considered noise of constraints. Sasena et al.¹⁵ suggested using a penalty added to the criteria, rather than multiplying the probability. This approach restricts sampling in the feasible region. Instead of the EI function, Gramacy and Lee¹⁶ proposed integrated expected conditional improvement and incorporated the constraint by setting a density function to the probability of satisfying the constraint. Additionally, a constrained optimization problem can be converted into unconstrained problems via augmented Lagrangian.^{17,18} The second way to treat constraints is to explicitly consider them in the optimization methodology. For example, the constraints can be modeled with kriging predictors and directly used as constraints in the infill sampling subproblem to filter out infeasible points.¹⁹ Some other forms of constraints for the subproblem have also been proposed, including expected violation of the constraints,²⁰ upper trust bound,²¹ and modified the probability of feasibility.²²

Black-box constraints can also be handled using surrogate-based feasibility analysis. First, the constraints can be consolidated into the concept of feasibility function $\psi(x)$, as shown in Equation (2). It measures the maximum violation of all the constraints. A set of inputs is only feasible when $\psi(x) \leq 0$.

$$\psi(x) = \max_{i \in I} \{g_i(x)\} \quad (2)$$

Feasibility analysis characterizes the feasible region in the input space. Surrogate-based approaches have been proposed for feasibility analysis. Such methods build surrogate models to approximate the feasibility function and exhibit a number of advantages. First, it does not require any underlying assumption about the characteristics of feasible region. Second, it can handle black-box constraints since only input x and output $\psi(x)$ are needed. Third, it requires small number of function evaluations, especially when facilitated with adaptive sampling strategies. Boukouvala and Ierapetritou²³ proposed a modified EI function for feasibility to balance the local search along feasibility boundary and global exploration. Wang and Ierapetritou²⁴ formally derived the EI function for feasibility based on a newly defined term called “improvement for feasibility.” Wang and Ierapetritou²⁵ used radial basis functions (RBF) as the surrogate for feasibility function approximation and proposed an indicator of uncertainty for RBF predictions. Metta et al.²⁶ investigated artificial neural network and proposed an uncertainty measurement based on Jack-knifing.

In this work, a new framework for feasibility-driven optimization is developed. First, we are taking a different perspective by viewing the feasibility analysis problem as a classification problem. Classification models, specifically, support vector machines (SVM) are used for feasibility characterization. Adaptive sampling strategy for SVM refinement near feasible region boundary is developed. This stage to determine the feasible space using SVM surrogate model is called “feasibility stage.” Second, SVM is incorporated as a constraint for the Gaussian process (GP)-based optimization problem, with the aim to locate the optimum within the feasible region. This forms the second

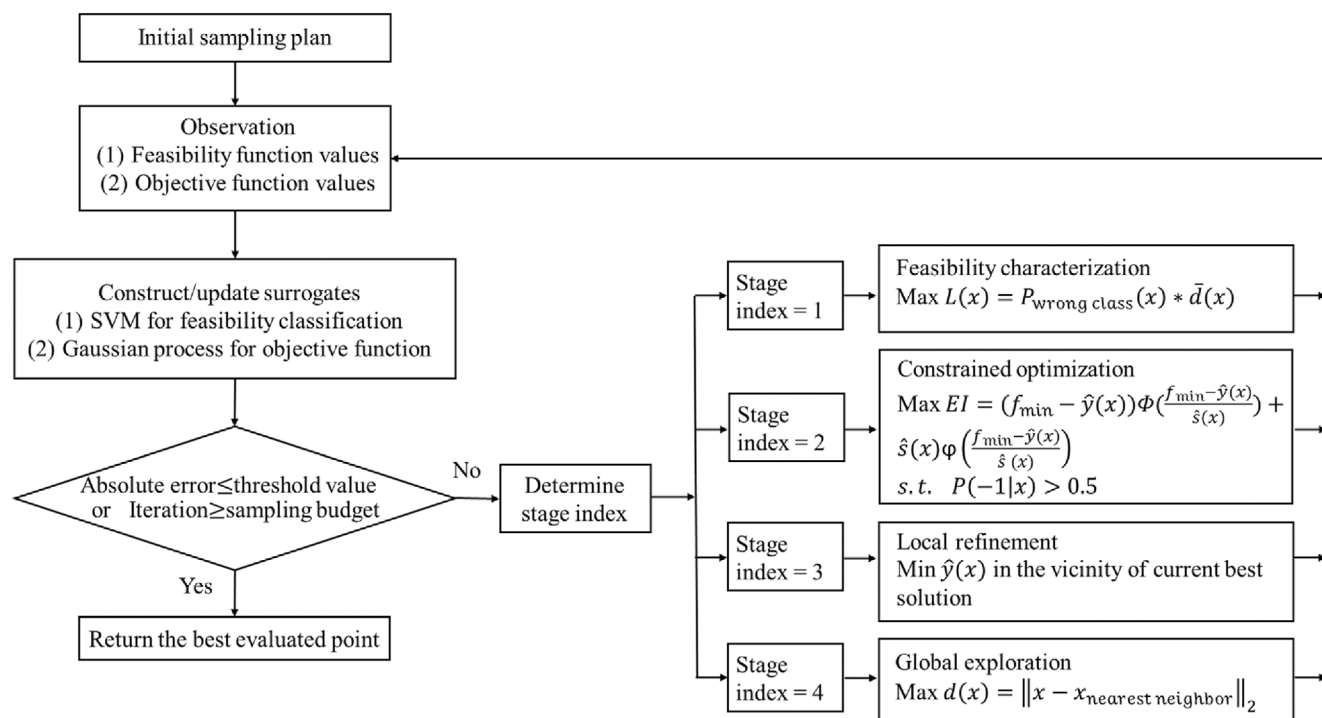


FIGURE 2 Flowchart of the three-stage dynamic switching surrogate-based optimization (SBO) framework.

stage, namely “optimization stage.” Third, a local refinement stage is introduced to refine local optimization and accurately locate the optimum. Finally, a global exploration stage is also added to improve global coverage. Switching strategies are developed to determine the stopping criteria for feasibility and optimization, local refinement, and global exploration stages. The outline of the paper is as follows. The surrogate-based feasibility analysis and optimization method is described in the next section, including the principles of GP and SVM, infill criteria for feasibility analysis, optimization, and global exploration, switching strategies, and the entire framework of the proposed method. Subsequently, we present the benchmark problems and performance measures that are used to compare different algorithms, followed by discussions on computational results. A pharmaceutical case study is also presented. Finally, the work is summarized in the last section.

2 | PROPOSED DYNAMIC SWITCHING SBO FRAMEWORK

The flowchart of the proposed framework is shown as Figure 2. First, the initial points are generated using Latin hypercube sampling (LHS). The value for feasibility function and objective function are obtained, and the initial surrogate models are constructed. Before the stopping criterion is met for the entire algorithm, it goes into the iterative procedure to adaptively add new points. Stage index has been assigned for each stage, where 1 corresponds to feasibility stage, 2 corresponds to optimization stage, 3 corresponds to local refinement stage, and 4 corresponds to global exploration stage. The determination for stage

index is built in the switching strategies. When the stopping criterion is satisfied, the best evaluated point is returned.

Below, we explain key components within the framework, including the surrogate models, infill criteria, and switching strategies. We first introduce the basics of surrogate models, which include GP regression for objective function and SVM for feasibility evaluation. Next, the infill criteria are explained in detail which are used to guide the search for new sample points in different stages. Subsequently, switching strategies that are used to determine stopping criteria for each stage are described. Finally, the full framework is summarized.

2.1 | Surrogate models

2.1.1 | GP regression/kriging

A GP is a collection of random variables, any finite number of which have a joint Gaussian distribution.²⁷ A GP is completely specified by its mean and covariance functions. We define the mean function $m(x)$ and the covariance function $k(x, x')$ of a real process $\hat{f}(x)$ as follows according to the work of Rasmussen and Williams²⁷:

$$m(x) = E[\hat{f}(x)] \quad (3)$$

$$k(x, x') = E[(\hat{f}(x) - m(x))(\hat{f}(x') - m(x'))] \quad (4)$$

and the GP is written as follows:

$$\hat{f}(x) \sim \text{GP}(m(x), k(x, x')) \quad (5)$$

The predictor $\hat{f}(x)$ is the predicted value for the objective function $f(x)$ in Problem 1. The mean function $m(x)$ reflects the expected function value at input x . The covariance function $k(x, x')$ models the correlation between the function values at different input points x and x' . The function $k(\cdot)$ is commonly called the kernel of GP. A sensible assumption for the kernel function is that closer points are expected to have more correlation and similarity than points further away from each other.

If we consider the simple special case where the observations are noise free, the predictive distribution at the new point X_* follows a normal distribution with the mean²⁷

$$m(X_*) + K(X_*, X)K(X, X)^{-1}(f - m(X)) \quad (6)$$

and variance²⁷

$$K(X_*, X_*) - K(X_*, X)K(X, X)^{-1}K(X, X_*) \quad (7)$$

where f is the training outputs, X_* is the new input point or test point, $K(X, X)$ is a n -by- n covariance matrix between all observed points, $K(X_*, X_*)$ is a 1-by-1 covariance matrix between the newly introduced point, $K(X_*, X)$ is a 1-by- n covariance matrix between the new input point and the already observed points, and $K(X, X_*)$ is a n -by-1 covariance matrix between the observed points and the new input point. Each element in the covariance matrix is defined as

$$K_{ij} = k(x_i, x_j) \quad (8)$$

Kriging is usually used in the literature interchangeably with GP. The use of the word “kriging” in spatial statistics has come to be synonymous with “optimally predicting” or “optimal prediction” in space, using observations taken at known nearby locations.²⁸ Below, we give a brief summary of the kriging model to be compared with the GP.

Kriging is a method of interpolation for random spatial processes. Suppose, we have a process $Z(\cdot)$, which is a random process that satisfies Equation (9).

$$Z(x) = \mu(x) + \delta(x) \quad (9)$$

where $\mu(x)$ is the deterministic mean structure, and $\delta(x)$ is the correlated error structure that follows a zero-mean stochastic process with known covariance function. The kriging predictor can be obtained as Equation (10), which is adapted from Cressie²⁸

$$\begin{aligned} p^*(X_*) &= K(X_*, X)K(X, X)^{-1}Z(X) + (1 - K(X_*, X)K(X, X)^{-1}1)\mu \\ &= \mu + K(X_*, X)K(X, X)^{-1}(Z(X) - 1\mu) \end{aligned} \quad (10)$$

The mean-squared prediction error adapted from Cressie²⁸ is

$$E(Z(X_*) - p^*(X_*))^2 = K(X_*, X_*) - K(X_*, X)K(X, X)^{-1}K(X, X_*) \quad (11)$$

Comparing Equations (3)–(5) and (9), it can be noticed that the kriging model has a similar underlying assumption as GP. They both model the output as a regression term plus an error term that has spatial correlation. Comparing Equations (6) and (7) and (10) and (11), the predicted mean and variance of GP are the same as the predictor and mean-squared error of kriging. So, kriging can be regarded as the noise-free GP model. They provide the same predictive mean and variance if the same mean function and covariance function are chosen. “GP model” is used throughout the article in later sections to refer to the model applied in the proposed framework.

2.1.2 | Parameter estimation and hyperparameter tuning for GP regression

In this study, the mean and covariance functions are determined by minimizing the mean squared error of the model prediction from the options listed in Tables 1 and 2.

2.1.3 | SVM model

SVM is a classification model that defines a hyperplane to separate samples of different classes. Given an input x , the output of SVM ($\hat{y}(x)$) is based on the labels of existing samples y_j and similarity between x and the existing samples x_j and is as follows:

$$\hat{y}(x) = \sum_j \hat{\alpha}_j y_j G(x, x_j) + \hat{b} \quad (12)$$

TABLE 1 Mean functions.

Constant	$m(x) = \beta_0$
Linear	$m(x) = x^T \beta_1 + \beta_0$
Quadratic	$m(x) = x^T \beta_2 x + x^T \beta_1 + \beta_0$

TABLE 2 Covariance functions.

Linear	$k(x, x') = \prod_{d=1}^D \max\{0, 1 - \theta_d x_d - x'_d \}$
Exponential	$k(x, x') = \prod_{d=1}^D \exp(-\theta_d x_d - x'_d)$
Gaussian	$k(x, x') = \prod_{d=1}^D \exp(-\theta_d x_d - x'_d ^2)$

Note: D is the dimension of x . θ_d are the determined within the range of [0.1, 10].

$$G(x_j, x_k) = \exp\left(-\frac{\|x_j - x_k\|^2}{\sigma^2}\right) \quad (13)$$

where $\hat{a}_j$ and $\hat{b}$ are model parameters to be fitted; $G(x_j, x_k)$ is the RBF kernel function which is used to transform the input space to a higher dimensional space where the samples are linearly separable. The output $\hat{\psi}(x)$ is the predicted value for the feasibility function $\psi(x)$ in Equation (2). It is called classification score, and it represents the distance x from the decision boundary. If $\hat{\psi}(x)$ is positive, x belongs to $+1$ class, if $\hat{\psi}(x)$ is negative, x belongs to -1 class, and if $\hat{\psi}(x)$ is zero, x lies on the decision boundary. One feature of SVM model is that the output of an SVM model is only dependent on a subset of samples, which are called support vectors. This is because only support vectors have nonzero corresponding $\hat{a}_j$ in Equation (12). For other samples x_j , the corresponding coefficients $\hat{a}_j$ are zero and thus $\hat{\psi}(x)$ will not be dependent on those x_j and y_j values. More intuitively, support vectors are the points closest to the opposite class. The SVM model is constructed by identifying these points to accurately separate different classes. This feature provides an intuitive justification for the infill criterion of feasibility stage, as explained in more detail in section 2.2.1.

2.1.4 | Parameter estimation and hyperparameter tuning for SVM

The hyperparameters considered in this study are the kernel scale σ in Equation (13) and the upper bounds of $\hat{a}_j$ parameter C in Equation (15). The values of hyperparameters have a significant impact on parameter estimation and final performance of the SVM model. Hyperparameter tuning is an optimization problem. The objective function is the loss function that computes the classification error of the model. There are different types of loss functions, such as misclassified rate, hinge loss, exponential loss.²⁹ Misclassified rate is used in this study.

The misclassified rate is defined as:

$$\text{Loss} = \sum_{j=1}^n w_j I\{\hat{y}_j \neq y_j\} \quad (14)$$

where w_j is the weight for observation j , and $I\{\cdot\}$ is the indicator function. Equal weights are assigned for each sample and the sum of weights is equal to 1, that is, $\sum_{j=1}^n w_j = 1$. Misclassified rate calculates the fraction of misclassified points. One consideration for the simulation data are that since the sample points are few and are obtained through expensive simulations, it is important to make sure that every training sample is classified correctly, which means $\text{Loss} = 0$. The second consideration is to avoid overfitting. A larger kernel scale σ is preferred to reduce the possibility of overfitting. To satisfy the two considerations above, the following steps are used to determine the hyperparameters. First, a pool of hyperparameters is generated by grid sampling in the hyperparameters space on log scale. The upper bound for both hyperparameters is 1000, and the lower bound is 0.001. One hundred (10×10) pairs of hyperparameters are generated in this study.

Next, L is calculated for each pair of hyperparameters, and the pairs with nonzero Loss values are screened out. Finally, the pair with maximum σ in the remaining pool is selected. In this way, the optimum hyperparameter values are determined.

After the hyperparameters are selected, the parameters $\hat{a}_j$ and $\hat{b}$ in Equation (12) are determined by maximizing the margin width of the separating hyperplane. The optimization formulation is shown in Equation (15). More details can be found by Hastie et al. and Cristianini and Shawe-Taylor.^{29,30} Parameter estimation is performed using sequential minimal optimization implemented in MATLAB R2020a using the “fitsvm” function, which is developed for training SVM models for binary classification.

$$\begin{aligned} \max_{\alpha} L &= \sum_j \alpha_j - \frac{1}{2} \sum_j \sum_k \alpha_j \alpha_k y_j y_k G(x_j, x_k) \\ \text{s.t. } \sum_j \alpha_j y_j &= 0 \\ 0 &\leq \alpha_j \leq C \end{aligned} \quad (15)$$

2.2 | Infill criteria

For surrogate-based feasibility analysis and optimization, the two key components are the surrogate model and the infill criteria. The purpose of the surrogate model is to build cheap models that approximate the original complex functions based on input and output data. The rule of the infill criteria is to quantify the utility at a point x (it is also called utility functions or acquisition functions in literature of Brochu et al.³¹). Thus, new valuable data points can be added iteratively to the database and the surrogate model will be updated and improved. In our framework, different infill criteria have been developed for different purposes: (1) feasibility learning function is used for feasibility characterization in the feasibility stage; (2) EI subject to feasibility constraint is used for objective improvement in the optimization stage; (3) surrogate predictor subject to box constraints in the vicinity of current best point is used for local refinement stage; (4) distance to nearest neighbor is used to sample the sparse region in the global exploration stage.

2.2.1 | Feasibility learning function maximization

In the feasibility stage, new sample points can be selected by maximizing the feasibility learning function as shown in Equation (16)³²:

$$\text{Max } L(x) = P_{\text{wrong class}}(x) * \bar{d}(x) \quad (16)$$

where $P_{\text{wrong class}}(x)$ is the probability of wrong classification, $\bar{d}(x)$ is the normalized distance to nearest neighbor. $P_{\text{wrong class}}(x)$ is defined as:

$$P_{\text{wrong class}}(x) = \min(P(-1|x), P(+1|x)) \quad (17)$$

where $P(-1|x)$ and $P(+1|x)$ are calculated based on the SVM prediction $\hat{\psi}(x)$, as shown below:

$$P(-1|x) = \frac{1}{2 + \hat{\psi}(x)}, \text{ if } \hat{\psi}(x) > 0 \quad (18)$$

$$P(-1|x) = \frac{1}{1 + \exp(\hat{\psi}(x))}, \text{ if } \hat{\psi}(x) < 0 \quad (19)$$

$$P(+1|x) = 1 - P(-1|x) \quad (20)$$

$P(-1|x)$ and $P(+1|x)$ are the probabilities that a point x is feasible and infeasible, respectively. A point that is highly likely to be feasible (in the feasible region and far away from the boundary) will have high $P(-1|x)$ value and low $P(+1|x)$ value; in contrast, a point that is highly likely to be infeasible (in the infeasible region and far away from the boundary) will have low $P(-1|x)$ value and high $P(+1|x)$ value. If a point has similar $P(-1|x)$ and $P(+1|x)$ values (i.e., both of them are close to 0.5), it is close to the SVM model boundary and is highly likely to be misclassified into the wrong class. Identifying the true class of this type of points is helpful for improving the accuracy of the SVM model. The sum of $P(-1|x)$ and $P(+1|x)$ is set to be 1. The range of $P_{\text{wrong class}}(x)$ is (0, 0.5).

Plus, $\bar{d}(x)$ is defined as:

$$\bar{d}(x) = \frac{d(x)}{\max(d(x))} \quad (21)$$

where $d(x)$ is the distance of point x to its nearest neighbor, $\max(d(x))$ is the distance of point x to its farthest neighbor.

To summarize, maximizing the feasibility learning function aims at selecting points with both high $P_{\text{wrong class}}(x)$ and $\bar{d}(x)$ value. More intuitively, a point is selected

1. If it is close to the current predicted boundary by SVM so it is likely to be misclassified.
2. If it is in the region where samples are sparse.

These two aspects are similar to the idea of balancing local exploitation and global exploration in many acquisition functions developed in the field of surrogate based optimization framework. A graphical illustration of the feasibility learning function can be found in Supporting Information Appendix A.

2.2.2 | EI maximization subject to feasibility constraint

After the SVM model is trained and improved in the feasibility stage, it is incorporated as constraint in the optimization stage to restrict the following search within the feasible region. The infill criterion is to maximize the EI function subject to the constraint that x is feasible:

$$\text{Max EI} = (f_{\min} - \hat{f}(x)) \Phi\left(\frac{f_{\min} - \hat{f}(x)}{\hat{s}(x)}\right) + \hat{s}(x) \phi\left(\frac{f_{\min} - \hat{f}(x)}{\hat{s}(x)}\right) \quad (22)$$

$$\text{s.t. } P(-1|x) > 0.5$$

where Φ is the normal cumulative distribution function, ϕ is the normal probability density function, $f_{\min}$ is the current best objective value found, $\hat{f}(x)$ is the predicted mean at point x , and $\hat{s}(x)$ is the standard deviation of the prediction.

2.2.3 | GP predicted mean minimization subject to feasibility constraint and box constraint

To further refine the optimal solution, the GP predicted mean is used as the objective function for a local search. The search is restricted inside a rectangle around the current best solution. The length of the side of the rectangle on dimension d is equal to 1/5 of the total length on dimension d , unless it exceeds the upper and lower bounds of parameter space. The complete infill criterion is shown in Equation (23).

$$\text{Min } \hat{f}(x)$$

$$\text{s.t. } P(-1|x) > 0.5$$

$$\max(x_d^{lb}, x_d^{\text{current best}} - \delta_d) \leq x_d \leq \min(x_d^{ub}, x_d^{\text{current best}} + \delta_d)$$

$$\delta_d = \frac{x_d^{ub} - x_d^{lb}}{10}, d \in \{1, 2, \dots, D\} \quad (23)$$

2.2.4 | Distance to nearest neighbor maximization

A pure global exploration stage is introduced to increase the possibility of finding the true optimum, and also as a guarantee for convergence at infinite number of sample points. The criterion used is the Euclidean distance to the nearest neighbor, as shown in Equation (24). By maximizing the distance, a point will be selected if it is in a region without many points.

$$\text{Max } d(x) = \|x - x_{\text{nearest neighbor}}\|_2 \quad (24)$$

2.2.5 | Optimization methods for the infill criteria

All the infill criteria optimization subproblems are solved using the interior point method implemented in Matlab R2020a using the “fmincon” function. Optimality tolerance is set as 1e-16.

2.3 | Switching strategies

With the feasibility stage, optimization stage, local refinement stage, and global exploration stage developed, the next question to ask is “how many points are needed for each stage.” The number of points

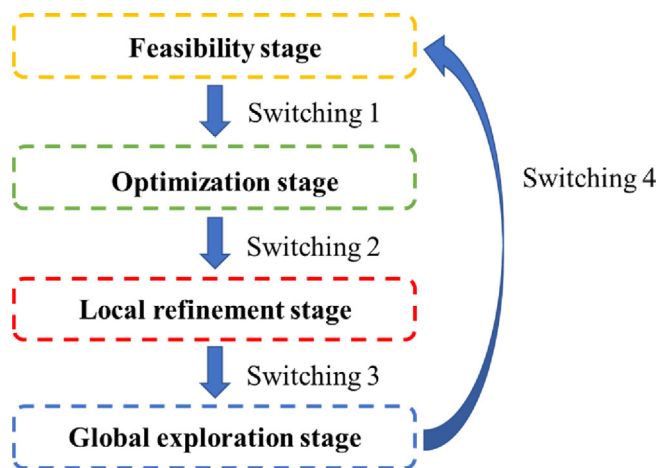


FIGURE 3 Schematic figure of the three-stage dynamic switching surrogate-based optimization (SBO) framework.

required for each stage depends on problem characteristics. For example, a problem with complex (e.g., nonconvex and disjoint) feasibility is highly likely to require more points to accurately characterize than simple problems. Similarly, a complex objective function might require more iterations to optimize. The interaction between the feasibility function and the objective function also matters. For example, if the optimum lies on the boundary or in a “small island” of the feasible region, it is important to accurately characterize the boundary and identify the “island” to find the optimum. The problem characteristics can be hard to access based on limited initial samples. Thus, it will be hard to assign sampling budget for each stage before the adaptive sampling procedure starts. A possible solution to facilitate this decision is to develop reasonable stopping criterion for each stage dynamically based on the performance of each stage, and ultimately when the entire algorithm stops, the total sampling budget has been reasonably distributed between different stages. As shown in Figure 3, we have set four switching strategies to switch from feasibility stage to optimization stage, optimization stage to local refinement stage, local refinement stage to global exploration stage, and finally global exploration stage back to feasibility stage.

Below, we listed the details for the switching strategies used for computational studies in section 3. Switching 1 indicates that the feasibility surrogate model is not changing significantly, so feasible region identified by initial points is likely to be accurately characterized and it is time to switch to optimization stage. Switching 2 indicates that the optimal solution found is not improving, so the algorithm should go to the next stage. Switching 3 means that the local refinement is complete. Switching 4 means that a new feasible point is found that has a promising objective value and also changes the feasibility surrogate (if the new point is a support vector, it is changing the formula of the SVM model). Such points most likely represent a newly discovered region, or represent a change in the boundary of an already discovered region, in which case we need to go back to the feasibility stage to sample the boundary again.

- Switching 1: For 10 consecutive iterations, the relative change in feasible area/volume based on the surrogate model is <5%.

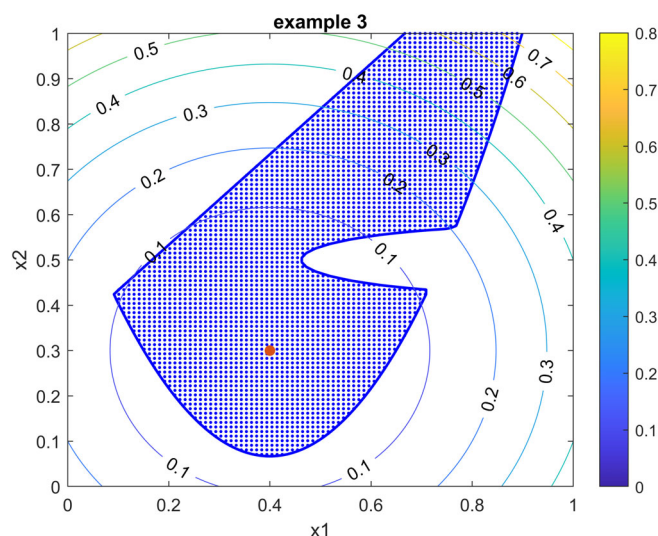


FIGURE 4 The ex3 problem. Feasible region is shaded with blue dots. Feasibility boundary is represented as the bold blue line. Objective function is depicted by the contour lines. Global optimum is shown as the red dot.

- Switching 2: For 10 consecutive iterations, the optimum found does not change.
- Switching 3: The algorithm has spent more than three iterations in Stage 3, or the new added sample is the same as existing samples.
- Switching 4: A new found point satisfies all the requirements below:
 - The point is feasible;
 - The point has an objective value that is less than the mean objective value of all the existing sample points;
 - The point is a support vector.

Note that the parameters in the switching strategies (i.e., 10, 3 iterations and 5% threshold value) are empirically determined based on a number of different test problems used.

3 | COMPUTATIONAL STUDIES

Three analytical problems are used to test the performance of the proposed algorithm. Detailed description for each problem is provided below, including the problem formulation, global optimum, and plots of feasibility and objective functions are shown in Figures 4–6.

Ex3 problem³³:

$$\begin{aligned}
 &\min (x_1 - 0.4)^2 + (x_2 - 0.3)^2 \\
 &\text{s.t. } -2x_1 + x_2 - 15 \leq 0 \\
 &\quad \frac{x_1^2}{2} + 4x_1 - x_2 - 5 \leq 0 \\
 &\quad -\frac{(x_1 - 4)^2}{5} - 2x_2^2 + 10 \leq 0
 \end{aligned} \tag{25}$$

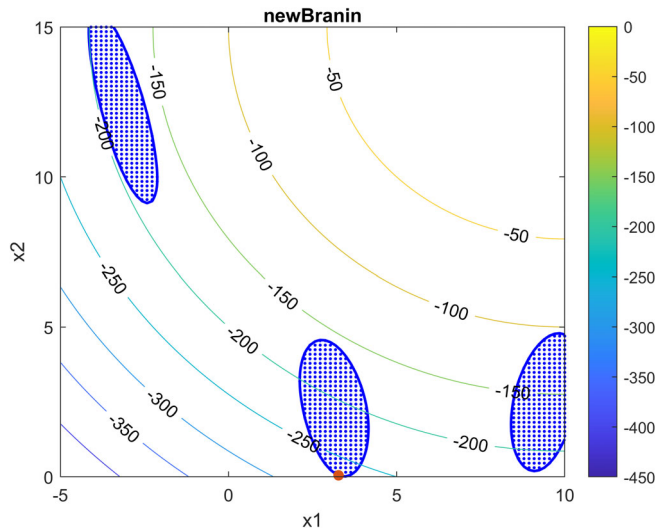


FIGURE 5 The new branin problem. Feasible region is shaded with blue dots. Feasibility boundary is represented as the bold blue line. Objective function is depicted by the contour lines. Global optimum is shown as the red dot.

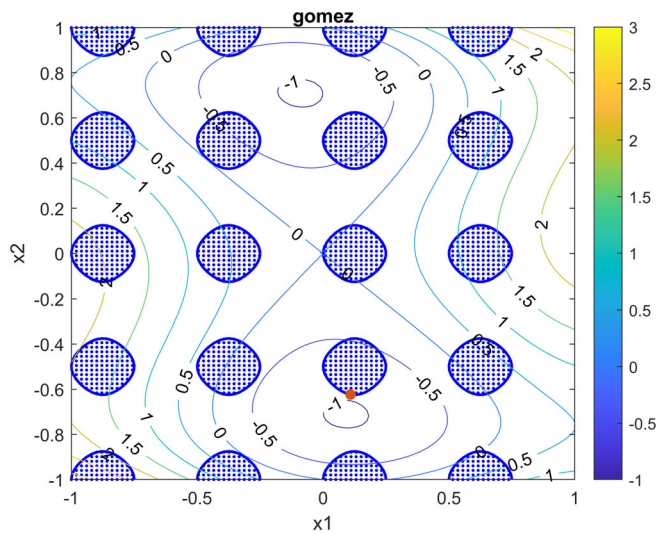


FIGURE 6 The gomez problem. Feasible region is shaded with blue dots. Feasibility boundary is represented as the bold blue line. Objective function is depicted by the contour lines. Global optimum is shown as the red dot.

Global optimum: $x_{\text{global}} = [0.4, 0.3]$, $y_{\text{global}} = 0$.

New branin problem³⁴

$$\min (x_1 + 3)^2 + (x_2 - 12)^2$$

$$\text{s.t.} \left(x_2 - \frac{5.1}{4\pi^2} x_1^2 + \frac{5}{\pi} x_1 - 6 \right)^2 + 10 \left(1 - \frac{1}{8\pi} \right) \cos x_1 + 5 \leq 0 \quad (26)$$

Global optimum: $x_{\text{global}} = [3.273, 0.0489]$, $y_{\text{global}} = -268.7879$.

Gomez problem³⁵:

$$\min \left(4 - 2.1x_1^2 + \frac{1}{3}x_1^4 \right) x_1^2 + x_1 x_2 + (-4 + 4x_2^2) x_2^2$$

$$\text{s.t.} -\sin(4\pi x_1) + 2(\sin(2\pi x_2))^2 \leq 0 \quad (27)$$

Global optimum: $x_{\text{global}} = [0.1093, -0.6234]$, $y_{\text{global}} = -0.9711$.

The number of initial points is chosen as 49 ($= 7^2$) for all problems in accordance with Wang and Ierapetritou.²⁵ The stopping criterion for the entire algorithm is as follows:

$$\text{Absolute value of relative error} = \left| \frac{f_{\min} - y_{\text{glob}}}{y_{\text{glob}}} \right| \leq 2\% \quad (28)$$

where $f_{\min}$ is the current best objective value found and y_{glob} is the true optimum of the objective function.

The algorithm is applied to the three test problems with 10 replications for each problem. For each replication, a new set of initial samples is generated based on LHS. The average number of iterations for different stages, total number of iterations, percentage of iterations for different stages across 10 runs for the test problems are listed in Table 3. The error plots showing the error as a function of iterations are included in the Supporting Information Appendix B. The performance comparison between the proposed approach and the two-stage GP-based approach is shown in the Supporting Information Appendix C.

The results above show that the algorithm is able to converge to the optimum with a small number of samples for all problems (on average 15.3 for ex3; 53.4 for new branin; 92.7 for gomez). This demonstrates the effectiveness of the proposed algorithm to solve problems with various characteristics. It should be noted that the number of iterations spent in each stage is problem dependent. The “ex3 problem” has a single large feasible region. The algorithm spends 14.3 iterations on average in feasibility characterization, which is the smallest among the three test problems. Then, it only needs 1 iteration to identify the optimum successfully since the objective function is a simple quadratic function. The algorithm does not go to the next stage since the absolute error stopping criterion is met after the optimization stage. This problem demonstrates the effectiveness of the proposed algorithm for problems with single nonconvex feasible region and convex objective function with the optimum lies inside the feasible region. The “new branin problem” has three medium-size feasible regions. The objective function is also a convex function, but the optimal feasible point lies on the boundary of the bottom middle region. For this problem, the algorithm spends more iterations (34.2) than ex3 problem in feasibility stage to acquire a representative surrogate model. Next, among the 10 replications, five runs are able to find the optimum at optimization stage without going to remaining stages. Four runs successfully find the optimum with the local refinement stage. One run goes to the global exploration stage and then restart from the feasibility characterization and terminates on optimization stage. This problem illustrates that disjoint feasible region requires more samples to characterize feasibility as expected and that for cases where the optimum is on the boundary, the local refinement stage may need for iterations to locate the optimum. This is because the

TABLE 3 Results for the test problems.

Problem	Average number of iterations spent in each stage and in total (percentage of iterations spent in each stage of total number of iterations)				
	Feasibility	Constrained optimization	Local refinement	Global optimization	Total
Ex3	14.3 (93.46%)	1 (6.54%)	0 (0%)	0 (0%)	15.3
New branin	34.2 (64.04%)	10 (18.73%)	1 (1.87%)	8.2 (15.36%)	53.4
Gomez	52.1 (56.20%)	26.6 (28.69%)	4.3 (4.64%)	9.7 (10.46%)	92.7

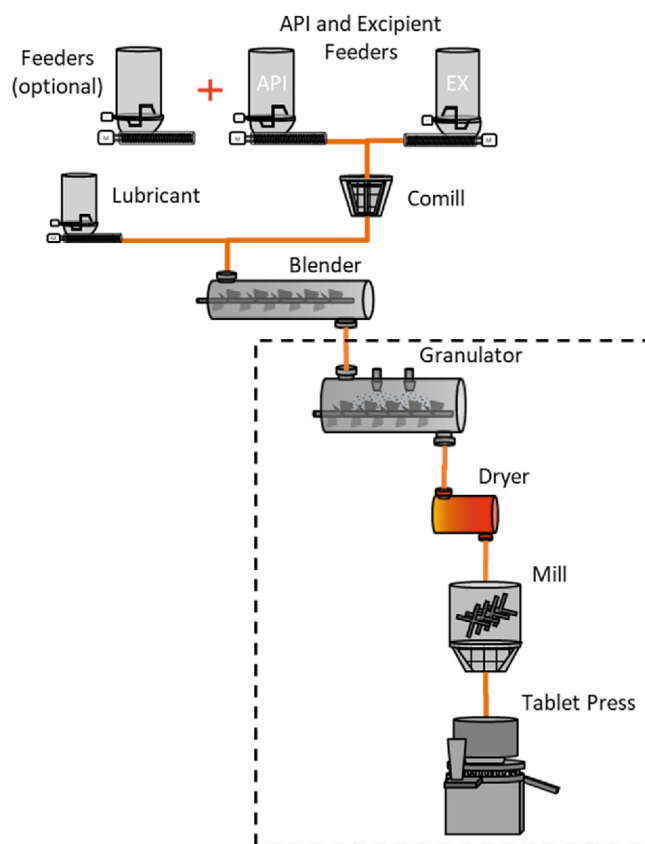
feasibility surrogate will limit the search within the feasible region, which might be smaller than the true feasible region, so the true optimum on the actual boundary is not included. A local refinement stage can then be used to drive the search towards the boundary by minimizing the predicted mean. The global exploration stage helps to identify disjoint regions when they are not found based on the initial sampling. The “gomez problem” has a number of small disjoint feasible regions, which leads to the largest number of iterations required for feasibility characterization. The optimum is located on the boundary of one small island of the feasible region. Among the 10 runs, one run is able to locate the optimum at the optimization stage without going to the local refinement and global exploration stages. Six runs utilize the global stage once to discover the small island containing the optimum, and then successfully locate in the following optimization stage. Two runs utilize the global stage twice to discover the small island, and one run utilizes it three times. In total, the gomez problem used more points than the other two problems for each stage.

4 | WET GRANULATION PHARMACEUTICAL MANUFACTURING PROCESS CASE STUDY

We also applied the algorithm to a real-world case study of a wet granulation process for solid-based drug manufacturing. As shown in the figure below, the active pharmaceutical ingredient and excipient are added using continuous feeders, followed by comill to de-lump the powders. Then lubricant is added to aid the flow of powders. The powders are mixed uniformly inside the blender, then the powder mixture goes through the granulator where liquid is added and the powder aggregates to granules. Then, a fluidized bed dryer is used to reduce moisture content to a required level. Next, mill is used to break granules to the required size. Finally, the granules are pressed into tablets. In this case study, a process simulation is developed from granulator to tablet press units, which is marked with a box with black dashed line in Figure 7. The simulation for the process is based on the population balance model, which is described in more detail by Metta et al.³⁶

The problem formulation is shown in Equation (29).

$$\min \text{Cost}_{\text{total}}(\mathbf{x}) = \text{Cost}_{\text{material}} + \text{Cost}_{\text{utility}}$$

**FIGURE 7** Flowsheet of the wet granulation process for continuous solid-based drug manufacturing.

$$\text{s.t. Yield}(\mathbf{x}) \geq 0.75$$

$$\text{Moisture content}(\mathbf{x}) \leq 1\%$$

$$\text{Total energy}(\mathbf{x}) \leq 5.3 \cdot 10^8 \text{ kJ/year} \quad (29)$$

where $\text{Cost}_{\text{material}}$ denotes total material cost for API, excipient, and lubricant; $\text{Cost}_{\text{utility}}$ denotes total utility cost; Yield denotes fraction of particles of acceptable size after co-mill; Moisture content denotes mass fraction of the liquid after dryer; Total energy denotes total energy consumed per year, $\mathbf{x}$ is the vector of input variables, including granulator speed (rpm), granulator liquid/solid ratio, dryer air velocity (m/s), dryer air temperature ($^{\circ}\text{C}$), and dryer dry time (min).

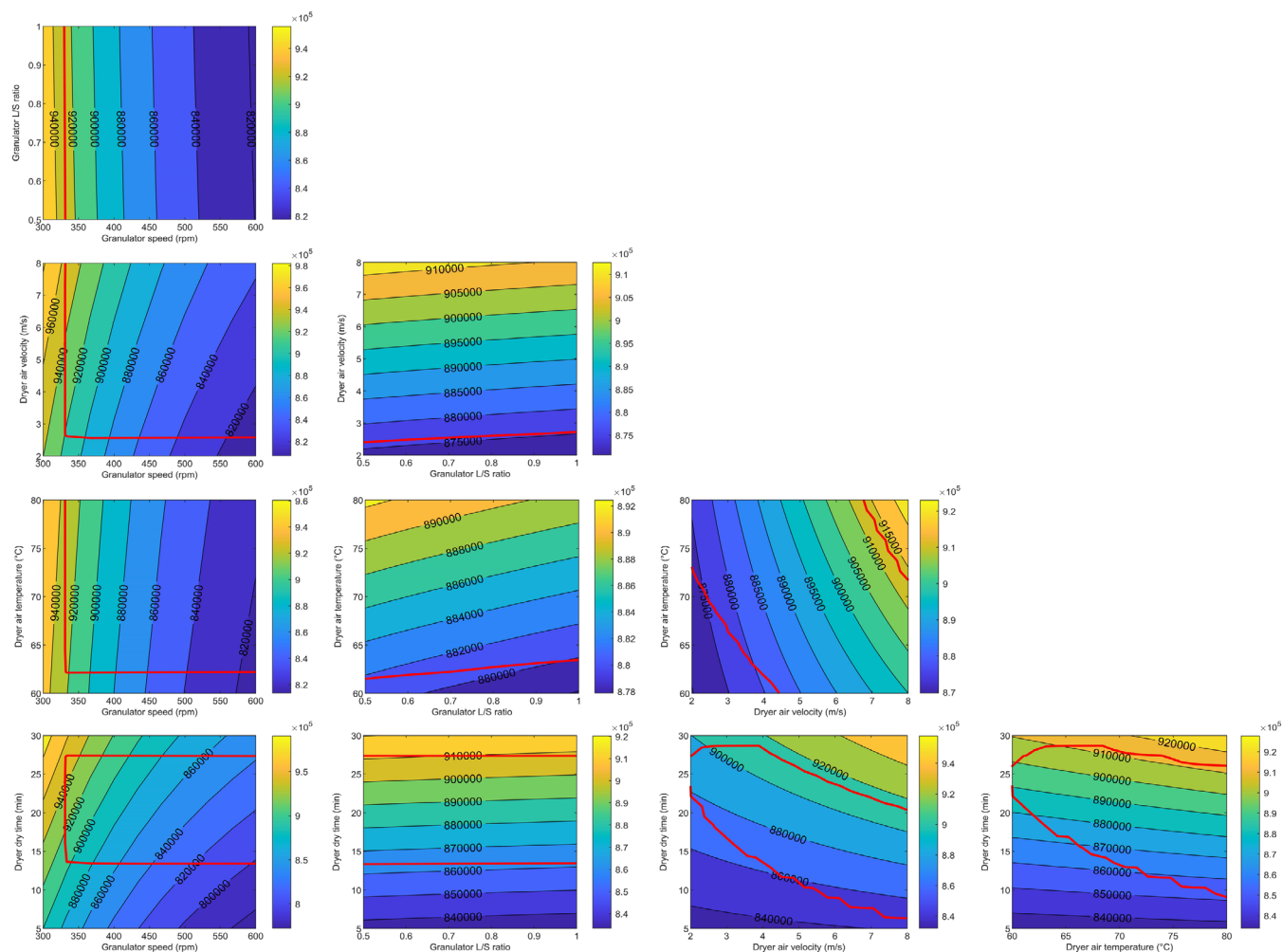


FIGURE 8 Feasibility boundary and objective function contour. Red solid line is the boundary of the feasible region. Contour lines show the objective function value.

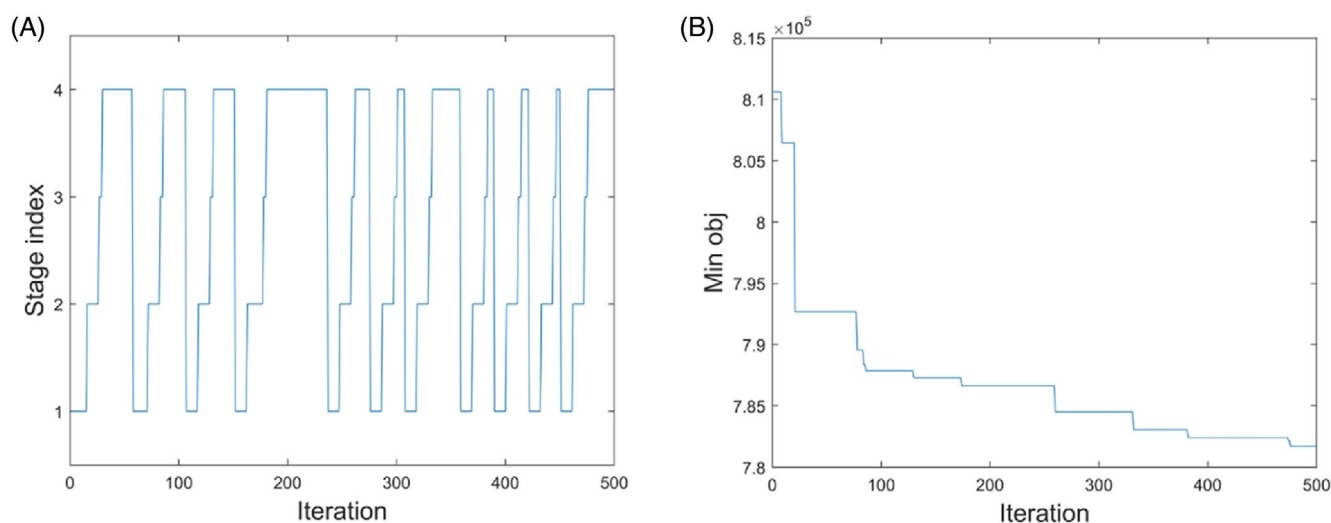


FIGURE 9 Results for the wet granulation process case study. (A) Stage index as a function of number of iterations. (B) Minimum objective function value as a function of number of iterations.

TABLE 4 Comparison between nominal points and optimal solution for the case study.

Input variables	Nominal values	Optimal solution		
		Proposed approach	Two-stage GP-based approach	LHS based non-adaptive approach
Granulator speed (rpm)	400	600	590.39	587.13
Granulator L/S ratio	0.66	0.93	0.64	0.57
Dryer air velocity (m/s)	4	7.71	7.46	6.86
Dryer air temperature (°C)	70	77	73.11	76.46
Dryer dry time (min)	20	5.64	7.40	7.44
Objective function value (\$/year)	885,600	781,700	790,700	803,830
CPU time	N/A	9.97 h	4.64 h	29.24 h

The feasible region of the input space and the objective function contour lines are shown in Figure 8. For each two-dimensional plot, the other input variables are set at the nominal values, as shown in Table 3.

To implement the proposed algorithm, the number of initial points is set as 50 and the stopping criterion for the entire algorithm is the number of iterations equal to 500.

The result is shown in Figure 9. The algorithm starts with the feasibility stage, and then switches between the four stages. The minimum objective value drops from 8.11×10^5 \$/year to 7.82×10^5 \$/year. The drops in Figure 9B happen at iteration 9, 21, 78, 84, 86, 130, 174, 260, 332, 382, 474, and 476, which corresponds to stage 1, 2, 2, 3, 4, 3, 2, 3, 3, 3, 3, and 4, respectively in Figure 9A. Most of the drops happen at stages 2 and 3, which matches the goals of the two designed stages. Some drops happen at stage 1 and 4, which is also possible since better solution can be found when characterizing the feasibility boundary or exploring the space.

As shown in Table 4, compared to the nominal value, the solution has a significant reduction of the operational cost. The total computation time (CPU time) is 9.97 h, from which 5.15 h is used to run the simulations corresponding to 51.56%. All the computations are performed on the computer with 11th Generation Intel Xeon W-1350 processor, 12 MB Cache, 6 Core, 3.3–5.0 GHz.

The problem is also solved using a two-stage GP-based feasibility and optimization framework.³⁷ The framework utilizes GP model for both feasibility and objective functions. The acquisition function for feasibility stage is EI for feasibility multiplied by probability that the objective function value is lower than the current minimum. The acquisition function for optimization stage is EI multiplied by the probability that the point is feasible. The number of initial sampling points is also set to 50, followed by 250 points for feasibility stage and 250 for optimization stage. The objective function value drops from 8.12×10^5 \$/year to 7.91×10^5 \$/year.

The two approaches are also compared to non-adaptive sampling approach. An LHS sampling is used to sample 2000 points in the input domain. Among 2000 samples, 913 samples are feasible, and the optimal objective value is 8.04×10^5 \$/year.

It can be noticed that the three approaches can all give better solution than the nominal point, with the proposed approach identifying the best objective function.

4.1 | Data availability and reproducibility statement

Detailed calculations for test problems results reported in Table 3 are tabulated in Data S1. MATLAB scripts to generate objective and feasibility function values in Figure 8 are available in Data S1. Numerical data from Figure 9 is tabulated in Data S1. Detailed results for Table 4 are available as .mat files in Data S1. All the results can be reproduced by running the MATLAB scripts provided in the “MATLAB scripts” folder.

4.2 | Summary and future work

In conclusion, a surrogate-based feasibility-driven optimization framework for expensive simulation is developed. The framework utilizes surrogate models to approximate original feasibility and optimization functions. SVM is used for feasibility function and GP is used for objective function. The surrogate models are updated through iterative adaptive sampling procedure. Different infill criteria are proposed for different stages of adaptive sampling, including the feasibility stage, optimization stage, local refinement stage, and global exploration stage. The algorithm shows good performance for three test problems and one real-life case study. In our future work, we are interested in performing sensitivity analysis on user-defined parameters. It will also be interesting to investigate the difference between classification model and regression model for feasibility function approximation.

AUTHOR CONTRIBUTIONS

Huayu Tian: Conceptualization (lead); data curation (lead); formal analysis (lead); investigation (lead); methodology (lead); software (lead); visualization (lead); writing – original draft (lead). **Marianthi Ierapetritou:** Conceptualization (supporting); data curation (supporting); formal analysis

(supporting); funding acquisition (lead); investigation (supporting); methodology (supporting); project administration (lead); resources (lead); software (supporting); supervision (lead); visualization (supporting); writing – original draft (supporting); writing – review and editing (lead).

ACKNOWLEDGMENTS

The authors would like to acknowledge financial support from US Food and Drug Administration (FDA) through grant FDABAA-20-00123.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Huayu Tian  <https://orcid.org/0000-0001-7424-6001>

REFERENCES

- Amaran S, Sahinidis NV, Sharda B, Bury SJ. Simulation optimization: a review of algorithms and applications. *4OR-Q J Oper Res*. 2014;12(4):301-333.
- Rios LM, Sahinidis NV. Derivative-free optimization: a review of algorithms and comparison of software implementations. *J Glob Optim*. 2012;56(3):1247-1293.
- Hooke R, Jeeves TA. "Direct Search" solution of numerical and statistical problems. *J ACM*. 1961;8(2):212-229.
- Nelder JA, Mead R. A simplex method for function minimization. *Comput J*. 1965;7(4):308-313.
- Torczon V. On the convergence of pattern search algorithms. *SIAM J Optim*. 1997;7(1):1-25.
- Holland JH. *Adaptation in natural and artificial systems: an introductory analysis with applications to biology, control, and artificial intelligence*. University of Michigan Press; 1975.
- Kirkpatrick S, Gelatt CD, Vecchi MP. Optimization by simulated annealing. *Science*. 1983;220(4598):671-680.
- Powell MJD. UOBYQA: unconstrained optimization by quadratic approximation. *Math Prog*. 2002;92(3):555-582.
- Gutmann H-M. A radial basis function method for global optimization. *J Glob Optim*. 2001;19:201-227.
- Jones DR, Schonlau M, Welch WJ. Efficient global optimization of expensive black-box functions. *J Glob Optim*. 1998;13(4):455-492.
- Mockus J. *Bayesian approach to global optimization: theory and applications*. Vol 37. Springer Science & Business Media; 2012.
- Schonlau M. *Computer experiments and global optimization*. Department of Statistics, University of Waterloo; 1997.
- Gardner J, Kusner M, Xu Z, Weinberger K, Cunningham J. Bayesian Optimization with Inequality Constraints. Paper presented at: International Conference on Machine Learning; Beijing, China. 2014.
- Gelbart MA, Snoek J, Adams RP. Bayesian optimization with unknown constraints. *arXiv Preprint:1403.5607*. 2014.
- Sasena MJ, Papalambros P, Goovaerts P. Exploration of metamodeling sampling criteria for constrained global optimization. *Eng Optim*. 2002;34(3):263-278.
- Gramacy R, Lee H. Optimization under unknown constraints. *arXiv:1004.4027*. 2010.
- Gramacy R, Gray G, Se LD, et al. Modeling an augmented Lagrangian for Blackbox constrained optimization. *arXiv:1403.4890v3*. 2015.
- Picheny V, Gramacy R, Wild S, Sebastien LD. Bayesian optimization under mixed constraints with a slack-variable augmented Lagrangian. *arXiv:1605.09466*. 2018.
- Sasena MJ, Papalambros PY, Goovaerts P. The use of surrogate modeling algorithms to exploit disparities in function computation time within simulation-based optimization. In: Cheng GD, Gu Y, Liu S, Wang Y, eds. *Proceedings of the 4th congress on structural and multidisciplinary optimization*. ISSMO; 2001.
- Audet C, Denni J, Moore D, Booker A, Frank P. A surrogate-model-based method for constrained optimization. 8th Symposium on multidisciplinary analysis and optimization. 2000.
- Priem R, Bartoli N, Diouane Y. On the use of upper trust bounds in constrained Bayesian optimization infill criteria. Conference on AIAA aviation 2019 forum. 2019.
- Bagheri S, Konen W, Allmendinger R, et al. Constraint handling in efficient global optimization. In *Proceedings of the genetic and evolutionary computation conference*; 2017:673-680.
- Boukouvala F, Ierapetritou MG. Derivative-free optimization for expensive constrained problems using a novel expected improvement objective function. *AIChE J*. 2014;60(7):2462-2474.
- Wang Z, Ierapetritou M. Surrogate-based feasibility analysis for black-box stochastic simulations with heteroscedastic noise. *J Glob Optim*. 2018;71(4):957-985.
- Wang Z, Ierapetritou M. A novel feasibility analysis method for black-box processes using a radial basis function adaptive sampling approach. *AIChE J*. 2017;63(2):532-550.
- Metta N, Ramachandran R, Ierapetritou M. A novel adaptive sampling based methodology for feasible region identification of compute intensive models using artificial neural network. *AIChE J*. 2020;67(2):e17095.
- Rasmussen CE, Williams CKI. *Gaussian processes for machine learning*. The MIT Press; 2005.
- Cressie N. The origins of kriging. *Math Geol*. 1990;22(3):239-252.
- Hastie T, Tibshirani R, Friedman J. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Second Edition ed. Springer; 2017.
- Cristianini N, Shawe-Taylor J. *An introduction to support vector machines and other kernel-based learning methods*. Cambridge University Press; 2000.
- Brochu E, Cora VM, de Freitas N. A tutorial on Bayesian optimization of expensive cost functions, with application to active user modeling and hierarchical reinforcement learning. *arXiv:1012.2599*. 2010.
- Song C, Shafieezadeh A, Xiao R. High-dimensional reliability analysis with error-guided active-learning probabilistic support vector machine: application to wind-reliability analysis of transmission towers. *J Struct Eng*. 2022;148(5):04022036.
- Rogers A, Ierapetritou M. Feasibility and flexibility analysis of black-box processes part 1: surrogate-based feasibility analysis. *Chem Eng Sci*. 2015;137:986-1004.
- Sasena MJ. *Flexibility and efficiency enhancements for constrained global design optimization with kriging: mechanical engineering*. University of Michigan; 2002.
- Gomez S, Levy AV. The tunnelling method for solving the constrained global optimization problem with several non-connected feasible regions. *Numer Anal*. 1982;34-47.
- Metta N, Ghijs M, Schäfer E, et al. Dynamic flowsheet model development and sensitivity analysis of a continuous pharmaceutical tablet manufacturing process using the wet granulation route. *Processes*. 2019;7(4):234.
- Wang Z, Escotet-Espinoza S, Singh R, Ierapetritou M. Surrogate-based Optimization for Pharmaceutical Manufacturing Processes. 27th European symposium on computer aided process engineering; 2017:2797-2802.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Tian H, Ierapetritou MG. A surrogate-based framework for feasibility-driven optimization of expensive simulations. *AIChE J*. 2024;70(5):e18364. doi:10.1002/aic.18364